

NANDHAKUMAR M

SENIOR .NET FULL-STACK DEVELOPER | TECHNICAL LEAD

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PROFESSIONAL SUMMARY

Senior .NET Full-Stack Developer and Technical Lead with **5 years of experience** architecting and delivering enterprise-scale supply chain, procurement, and business intelligence platforms using ASP.NET Core, ASP.NET MVC, C#, .NET Core Web API, and SQL Server. Led a four-phase enterprise security remediation program that resolved **500+ OWASP Top 10 vulnerabilities** across a live production platform, and built SAP synchronization services from scratch that eliminated manual cross-team reconciliation. Proven in owning zero-downtime production releases on IIS with no SLA breaches, tuning high-volume T-SQL workloads to sub-second response times, and mentoring developers on secure coding and .NET best practices. Alongside enterprise work, delivers freelance products end to end — an offline-first ASP.NET Core 8 order management platform with Firestore sync, plus cross-platform Flutter and PWA applications shipped to Android. Skilled in AI-assisted development using Claude Code and ChatGPT to accelerate delivery and strengthen code review quality.

TECHNICAL SKILLS

Backend: C#, .NET Core, ASP.NET Core 8, ASP.NET MVC 5, .NET Core Web API, REST APIs, Entity Framework Core 8, Entity Framework 6, LINQ, ADO.NET, Dependency Injection, ASP.NET Identity, ASP.NET Core Identity, OWIN, Swagger / OpenAPI, API Versioning

Frontend: JavaScript, TypeScript, jQuery, Bootstrap, HTML5, CSS3, Razor Views, Responsive Web Design, Progressive Web Apps (PWA)

Mobile & Cross-Platform: Flutter, Dart, Capacitor 6, Android APK Build & Signing, Offline-First Architecture, Local Notifications, Google Sign-In

Cloud & Backend Services: Google Cloud Firestore, Firebase, Google Drive API, GitHub Pages, Firestore Security Rules

Database: Microsoft SQL Server, T-SQL, Stored Procedures, Query Optimization, Indexing Strategies, Execution Plan Analysis, Backup & Recovery

Security: OWASP Top 10, JWT Authentication, Role-Based Access Control (RBAC), XSS Prevention, CSRF Protection, CSP Headers, Input Validation, Parameterized Queries, DOMPurify, Secure File Upload, PBKDF2, AES-256, TLS 1.2+ Enforcement, Path Traversal & IDOR Prevention, Secure Configuration Management

DevOps & Deployment: Git, TFS, Azure DevOps, IIS Administration, Hangfire, Zero-Downtime Deployments, Postman, ELMAH, log4net, Production Support

Integrations & Document Processing: SAP Integration, RESTful Services, SMTP / Notification Services, Excel Bulk Data Pipelines, iTextSharp, Ghostscript, ClosedXML, Background Job Scheduling

Reporting & BI: DevExpress, DevExtreme, Crystal Reports, KPI Dashboards, Drill-Down Data Visualization, SQL Aggregation & Analytics

Leadership & Practices: Technical Mentoring, Code Review, Secure Coding Standards, Requirement Analysis, Release Management, Stakeholder Coordination

AI Tools: Claude Code, ChatGPT, Prompt Engineering, AI-Assisted Code Review & Development

PROFESSIONAL EXPERIENCE

Senior .NET Full-Stack Developer & Technical Lead

Apr 2023 – Present

Spark IT Tech, Tiruppur, Tamil Nadu

- Lead the technical direction of the enterprise supply chain and procurement platform, owning architecture decisions, release planning, and production stability across multiple concurrent modules.
- Directed a four-phase enterprise security remediation program that identified and resolved **500+ vulnerabilities** spanning XSS, SQL Injection, CSRF, and unsafe file upload paths on a live production system.
- Established secure coding standards for the development team and conduct recurring code reviews and security audits to enforce them across all new development.
- Mentor junior developers on .NET best practices, defensive coding, SQL performance tuning, and AI-assisted development workflows.
- Introduced AI-assisted development practices using Claude Code and ChatGPT across the engineering team, improving feature turnaround and code review depth.
- Own end-to-end IIS deployments, server configuration, and zero-downtime production releases for procurement and finance systems with **no SLA breaches**.
- Coordinate with business stakeholders to translate procurement, finance, and inventory requirements into scalable technical designs and delivery timelines.

.NET Full-Stack Developer

Aug 2021 – Mar 2023

Spark IT Tech, Tiruppur, Tamil Nadu

- Built and shipped core procurement and inventory management modules supporting high-volume daily transactions on the enterprise SCM platform.
- Developed RESTful .NET Core APIs with JWT authentication, API versioning, and Swagger documentation consumed by internal web and integration clients.
- Optimized complex T-SQL stored procedures and indexing strategies to achieve **sub-second response times** under peak transactional load.

- Designed Excel bulk upload modules with multi-layer validation, tooltip-driven templates, and SAP synchronization override logic, cutting operational onboarding time.
- Configured Hangfire recurring jobs with retry policies, queue management, and monitoring dashboards to move long-running workflows off the request path.

Freelance Full-Stack Developer

Sep 2025 – Present

Self-Employed · Remote

- Deliver client and product work end to end — requirement gathering, architecture, development, Android packaging, and deployment.
- Built an offline-first order and inventory management platform on ASP.NET Core 8 with a Firestore sync relay, letting field staff capture orders with no connectivity.
- Shipped cross-platform applications from single codebases using Flutter and Capacitor, published as installable PWAs and Android APKs.
- Own the full release path including signed Android builds, Firestore security rules, and static hosting with SEO and PWA manifests.

KEY PROJECTS

Supply Chain & Procurement Management System — Enterprise Internal Platform

Technologies: C#, ASP.NET MVC, .NET Core, Web API, Entity Framework, SQL Server, Hangfire, JWT, IIS, Azure DevOps, Git

- Architected and delivered a multi-module platform covering procurement, inventory, and supply chain operations for high-volume daily transactions.
- Designed the RESTful API layer with JWT authentication, versioning, and Swagger documentation for internal and integration consumers.
- Enforced 150-character validation rules consistently across Razor views, APIs, service layers, and Excel workflows, shipped with **zero production defects**.
- Tuned stored procedures, query plans, and indexes to sustain sub-second response times as transaction volume scaled.

Smart Documents — AP Invoice & Document Management Platform — Full Stack Developer

Technologies: C#, ASP.NET MVC 5, ASP.NET Web API, .NET Framework 4.8, Razor, Entity Framework 6 (Database First / EDMX), ADO.NET, SQL Server, Stored Procedures, ASP.NET Identity, OWIN / Katana, DevExtreme, jQuery, iTextSharp, Ghostscript.NET, ELMAH, log4net

- Built a **metadata-driven document type engine** where grid columns, captions, formatting, summary aggregations, hierarchical filters, and role-based access are configured as data — onboarding new document types without code changes.
- Implemented **Active/Archive data tiering** with runtime database switching, resolving filter master data from the live database while serving historical reads from the archive.
- Delivered SAP Business One integration covering invoice and vendor master data, including a token-authenticated API allowing SAP to attach documents directly against a document key.
- Engineered a secure document viewer using AES-encrypted URL parameters and an external-site allow list, so document links could be shared across systems without exposing internal keys.
- Designed a compressed per-session result cache that served DevExtreme paging and sorting from memory instead of re-querying SQL, alongside stored-procedure aggregation for dashboard counts.
- Built a full-request audit subsystem capturing user, action, client, geolocation, payload sizes, and response timings by wrapping the response stream in a global action filter.
- Implemented server-side PDF compression on an iTextSharp and Ghostscript subprocess pipeline with configurable quality thresholds and correct process, stream, and handle management.
- Delivered a **65-issue code review remediation** across 37+ files — 13 critical, 17 high, 35 medium — resolving 29 security, 13 performance, and 12 resource-management defects to a clean zero-error build, and eliminating N+1 query patterns for a **~50% reduction in queries**.

Enterprise Security Remediation Initiative — Lead Developer, Security | 4-Phase Program

Technologies: ASP.NET Core, C#, SQL Server, DOMPurify, CSP Headers, OWASP Top 10, JWT, RBAC, Parameterized Queries

- Led a four-phase remediation program across a live enterprise SCM platform, resolving **500+ vulnerabilities** without disrupting business operations.
- Applied DOMPurify sanitization, parameterized queries, and Content Security Policy headers to close XSS and SQL Injection attack surfaces.
- Implemented RBAC, XSS prevention filters, CSRF protection, and centralized input validation across every application entry point.
- Ran code reviews and security audits and codified the resulting secure coding standards for ongoing team use.

SAP Synchronization & Integration Services

Technologies: C#, .NET Core, Web API, Hangfire, SAP Integration, SQL Server, SMTP

- Built SAP synchronization services from scratch using Hangfire for near real-time procurement and finance data exchange.
- Eliminated manual cross-team reconciliation by automating bidirectional master and transactional data flows.
- Added retry policies, failure alerting, and queue monitoring so integration failures surface and self-recover without manual intervention.

BI Reporting & KPI Dashboard — Oct 2023 – Jan 2024

Technologies: C#, ASP.NET MVC, SQL Server, DevExpress, Crystal Reports, JavaScript, jQuery, Bootstrap

- Engineered real-time BI dashboards and KPI reporting modules using DevExpress controls and advanced SQL aggregation queries.
- Built Crystal Reports integrations for both scheduled and on-demand executive reporting.
- Designed drill-down data visualization layers giving stakeholders actionable insight into procurement and supply chain KPIs.
- Used Claude Code and ChatGPT to compress feature development and code review cycles.

Excel Bulk Upload & Data Validation Framework

Technologies: C#, ASP.NET MVC, .NET Core, SQL Server, SAP Integration, JavaScript

- Designed a reusable bulk upload framework with template generation, tooltip comments, and row-level validation feedback.
- Implemented SAP synchronization override logic to reconcile bulk-loaded records against master data safely.
- Reduced operational onboarding effort for high-volume master data entry across procurement modules.

FREELANCE PROJECTS

E-Palace Crackers — Order & Inventory Management Platform — Sep 2025 – Jul 2026

Technologies: ASP.NET Core 8 MVC, C#, Entity Framework Core 8, SQL Server, ASP.NET Core Identity, Google Cloud Firestore, ClosedXML, iTextSharp, PdfSharp

- Built a full order management platform for a Sivakasi fireworks business covering customers, item and stock master, order headers and details, order tracking, follow-ups, and payment modes.
- Designed an **offline-first sync architecture** keeping SQL Server as the master and Firestore as a relay, so field staff capture orders on an Android app with no connectivity and queue them on-device.
- Made synchronization **idempotent** using device-generated source IDs, so re-imported orders are skipped while invoice numbers, stock adjustments, and payment status update correctly.
- Abstracted the sync store behind an interface with a local file backend, allowing full development and testing without a cloud account.
- Delivered PDF invoice generation, Excel export, reporting dashboards, and email notifications, plus a separate SEO-ready static marketing site with PWA manifest and sitemap.

Family Finance — Cross-Platform Household Finance Manager — Jun 2026 – Jul 2026

Technologies: Flutter, Dart, Google Cloud Firestore, Google Drive API, Google Sign-In, Local Notifications, Firestore Security Rules

- Developed a Flutter application targeting **Android, iOS, and Web from one codebase**, managing salary, categorised expenses, EMIs, budgets, reminders, and reports.
- Implemented a shared **family common wallet** with head and member roles, family-code invitations, and per-member spend analytics.
- Built a dual storage backend — Excel workbooks in Google Drive with automatic switching to Firestore — so the app works before any cloud setup and scales once configured.
- Solved multi-device write conflicts by merging the latest remote copy before every save and tracking removals in a dedicated sheet so deletions propagate correctly.
- Added EMI payoff projection, per-category budget overspend warnings, and recurring reminders that fire as real device notifications even when the app is closed.

PixelKeep — Offline Image & PDF Compressor — Jul 2026

Technologies: JavaScript, Capacitor 6, Canvas API, pdf.js, jsPDF, JSZip, Android, PWA

- Built a compressor that hits an **exact target file size** by binary-searching encoder quality, landing just under the target at the highest achievable quality.
- Preserved original resolution by default so no pixels are dropped, applying downscale only as a fallback when a target is unreachable at full size.
- Implemented multi-format conversion across JPG, PNG, WebP, PDF, and DOCX, assembling the OOXML Word package entirely in-browser with JSZip.
- Shipped one codebase as both an installable PWA and a signed Android APK via Capacitor, with native filesystem save and share integration.
- Designed it to run **100% offline** with no network calls, uploads, or tracking, so files never leave the device.

KEY ACHIEVEMENTS

- Resolved **500+ security vulnerabilities** across a live enterprise SCM platform, aligning the application with OWASP Top 10 standards.
- Built SAP integration workflows from scratch, removing recurring cross-team reconciliation effort and enabling near real-time synchronization.
- Delivered measurable SQL performance gains on high-volume transactional modules through stored procedure and indexing optimization.
- Led zero-downtime production deployments for procurement and finance systems with **no SLA breaches**.
- Introduced AI-assisted development workflows within the engineering team, improving delivery speed and code quality.

EDUCATION

Bachelor of Engineering (B.E.) – Electrical & Electronics Engineering

2016 – 2020

Karpagam College of Engineering

- CGPA: 7.5 / 10

PROFESSIONAL DEVELOPMENT

Actively expanding into React.js, Azure Cloud, Docker, Microservices Architecture, and CI/CD pipelines to broaden modern full-stack and cloud-native delivery capability.

LANGUAGES

- Tamil (Native / Bilingual Proficiency)
- English (Professional Working Proficiency)